

Mini Assignment

HCI & Computer Graphics in Google Earth

Interactive 3D Software Analysis

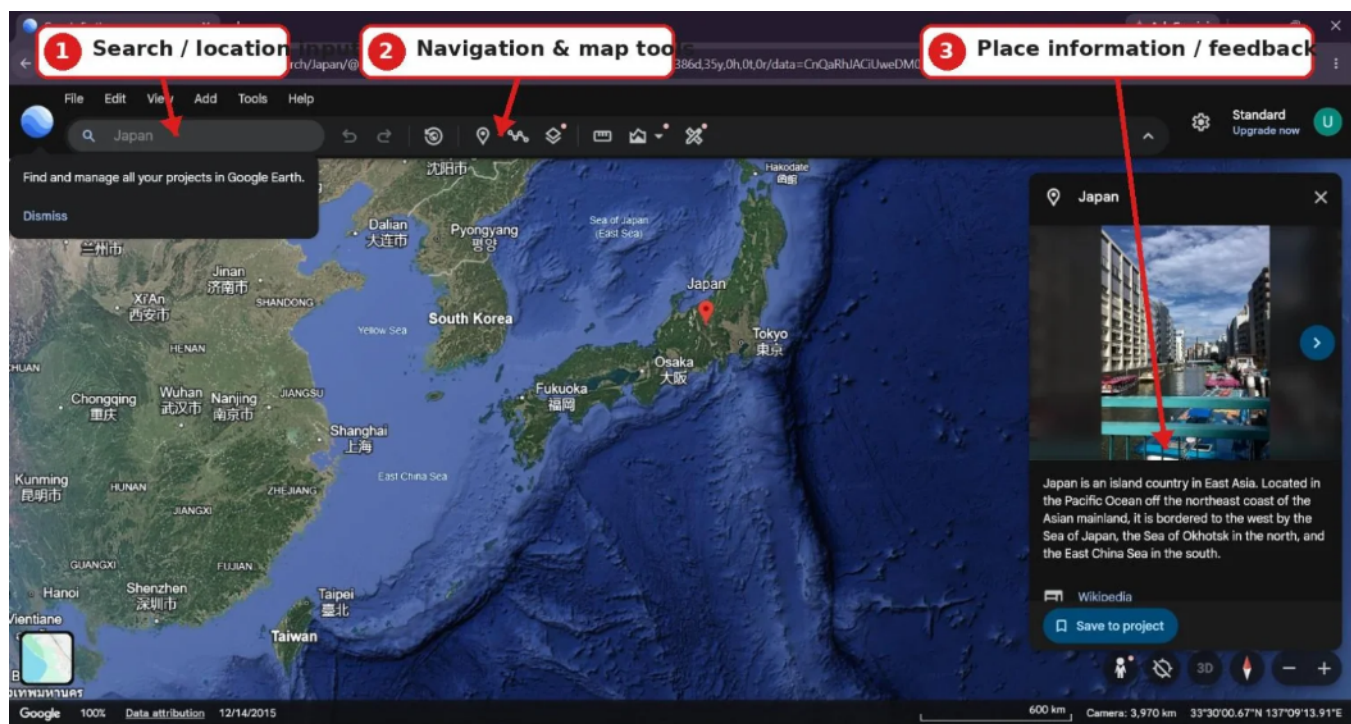
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ROLL NO.	2K24/CSE/85
CLASS / SEMESTER	6TH PRE-ENGINEERING
COURSE	HCI & COMPUTER GRAPHICS

I. SELECTED APPLICATION: GOOGLE EARTH

Google Earth is an interactive 3D geographical application that allows users to explore the Earth using satellite imagery, maps, 3D buildings, and other visual information. It demonstrates both Human-Computer Interaction (HCI) and Computer Graphics through its interface, navigation, 3D visualization, textures, and perspective.

2. SCREENSHOT I — HCI FOCUS

The screenshot below identifies three interaction/UI elements. The red numbered callouts point directly to each element.



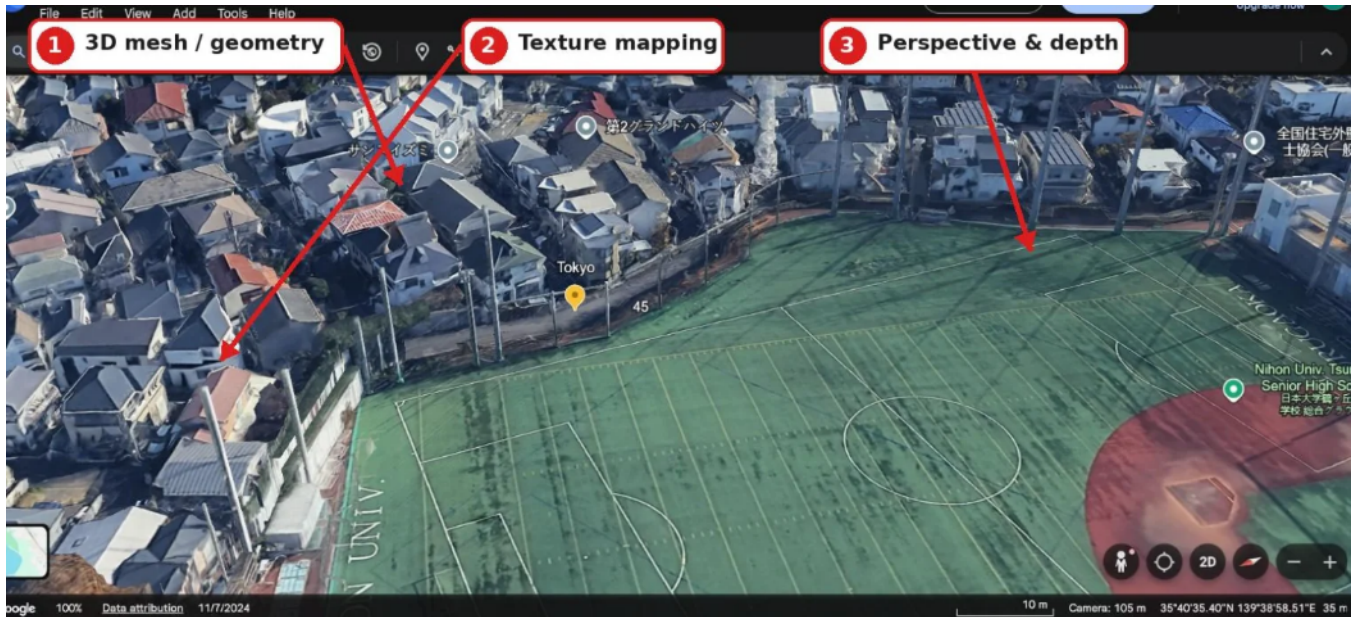
① SEARCH / LOCATION INPUT: ALLOWS THE USER TO ENTER A PLACE AND QUICKLY NAVIGATE TO IT.

Navigation & map tools: PROVIDE CONTROLS FOR NAVIGATION, VIEWING, ADDING INFORMATION, AND MAP.

Place information / feedback: DISPLAYS INFORMATION ABOUT THE SELECTED LOCATION, GIVING THE USER FEEDBACK AFTER AN INTERACTION.

3. SCREENSHOT 2 — COMPUTER GRAPHICS FOCUS

This screenshot shows three graphics/rendering elements used to create a 3D environment.



3D mesh / geometry: BUILDINGS AND OTHER STRUCTURES ARE REPRESENTED WITH THREE-DIMENSIONAL GEOMETRY THAT GIVES THEM SHAPE AND DEPTH.

Texture mapping: REAL-WORLD IMAGERY AND SURFACE DETAILS ARE MAPPED ONTO OBJECTS AND TERRAIN, MAKING THE SCENE MORE REALISTIC.

Perspective & depth: NEARBY OBJECTS APPEAR LARGER THAN DISTANT OBJECTS, HELPING CREATE THE VISUAL PERCEPTION OF 3D SPACE.

4. SHORT REFLECTION

1. How does the Computer Graphics quality help or hinder your ability to interact with the software? The realistic 3D graphics make Google Earth easier and more interesting to explore because users can understand locations through depth, perspective, and detailed textures. 3D buildings and satellite imagery help users recognize places more easily. However, highly detailed graphics may require more processing power and can make navigation slower on weaker devices.

2. What is one simple HCI change you would make to improve the user interface? I would make the navigation controls more customizable so users can choose which controls are displayed. This could reduce screen clutter and make the interface easier for beginners to use.

5. CONCLUSION

Google Earth demonstrates how HCI and Computer Graphics work together in interactive software. Its interface provides intuitive controls for searching and navigating, while 3D geometry, textures, perspective, and rendering create an understandable and realistic geographical environment.